

VITALCARE HEALTH SOLUTIONS PHISHING EMAIL ANALYSIS

TECHNICAL REPORT

GROUP 1

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1. Executive Summary

On March 6, 2026, VitalCare Health Solutions' security monitoring systems identified a targeted spear-phishing attack directed at internal staff. The attacker impersonated the internal security team to create a high-pressure scenario involving "unusual account activity" and the threat of immediate account suspension.

The primary objective of the attack was credential harvesting via a malicious external link disguised as an account verification link. While the email was professionally crafted to bypass basic human scrutiny, technical header analysis revealed that the email authentication protocols (SPF, DKIM, and DMARC) all failed, indicating the email did not originate from an authorized source. VirusTotal analysis of the embedded link showed that 5 out of 95 security vendors flagged the URL as malicious. When opened in a controlled sandbox environment, the link is redirected to a domain unrelated to the organization. These combined findings strongly suggest a targeted phishing attempt.

Timely intervention prevented unauthorized access to VitalCare's cloud-based Electronic Health Records (EHR) and patient management platforms, safeguarding sensitive healthcare data and ensuring HIPAA compliance. However, the incident highlights the importance of continued email authentication, awareness training, and use of security tools to mitigate these threats.

2. Phases of Analysis

2.1. Email Detection and Initial Review

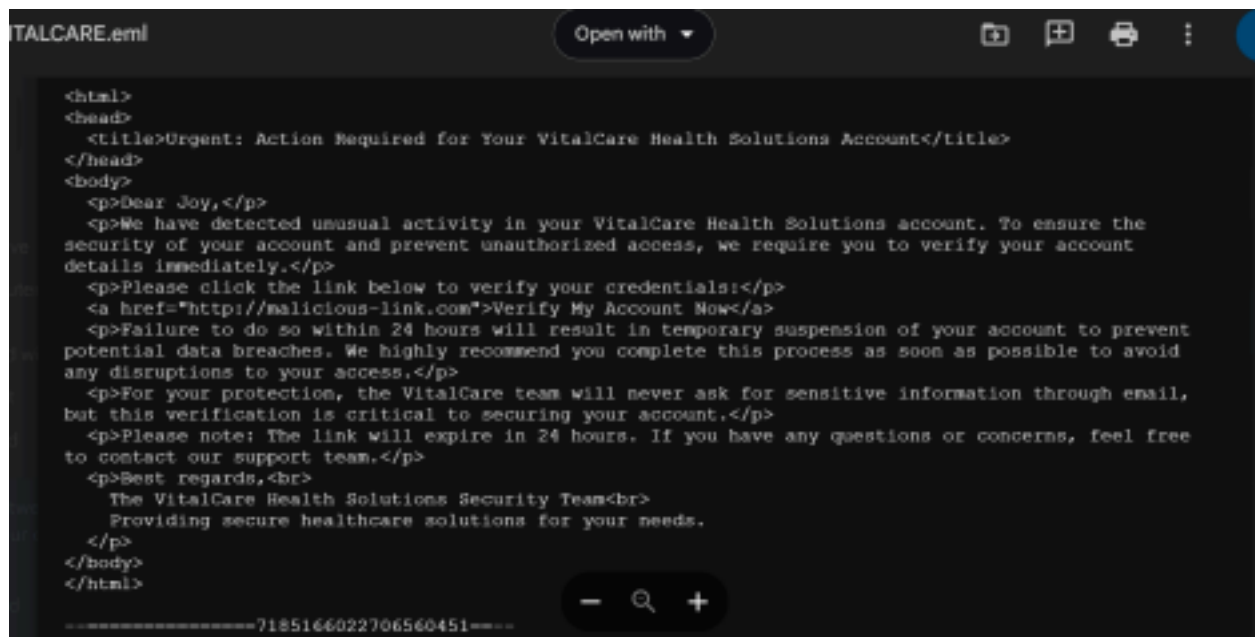
The initial review of the email reveals several Indicators of Compromise associated with phishing attempts.

- **Urgency:** The email subject, "*Urgent: Action Required for Your VitalCare Health Solutions Account*", employed urgency to prompt immediate action. The message claims that unusual activity has been detected on the recipient's VitalCare Health Solutions account and urges immediate action to verify account details. Such time pressure is a

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common social engineering tactic designed to prompt users to act quickly without verifying the legitimacy of the request.

- **Fear:** The email also uses fear by threatening that failure to act within 24 hours will result in temporary account suspension.
- **Building trust:** The email contains a statement claiming that the VitalCare team will never ask for sensitive information through email, yet it is urging the recipient to verify their account through the link in the email. This inconsistency is suspicious and is often used to create a sense of legitimacy and trustworthiness, making the message appear security conscious and encouraging the recipient to comply with the request.
- **Suspicious Hyperlink:** Additionally, hovering over the verification link reveals a suspiciously unrelated link that likely redirects users to a fake website rather than the legitimate VitalCare platform.



```
<html>
<head>
  <title>Urgent: Action Required for Your VitalCare Health Solutions Account</title>
</head>
<body>
  <p>Dear Joy,</p>
  <p>We have detected unusual activity in your VitalCare Health Solutions account. To ensure the security of your account and prevent unauthorized access, we require you to verify your account details immediately.</p>
  <p>Please click the link below to verify your credentials:</p>
  <a href="http://malicious-link.com">Verify My Account Now</a>
  <p>Failure to do so within 24 hours will result in temporary suspension of your account to prevent potential data breaches. We highly recommend you complete this process as soon as possible to avoid any disruptions to your access.</p>
  <p>For your protection, the VitalCare team will never ask for sensitive information through email, but this verification is critical to securing your account.</p>
  <p>Please note: The link will expire in 24 hours. If you have any questions or concerns, feel free to contact our support team.</p>
  <p>Best regards,<br>
    The VitalCare Health Solutions Security Team<br>
    Providing secure healthcare solutions for your needs.
  </p>
</body>
</html>
```

7185166022706560451

Fig1: Email body showing the Indicators of Compromise phishing

2.2. Email Header and Authentication Analysis

Analysis of the email headers revealed several red flags:

- **SPF (Sender Policy Framework): Fail**, The sending IP (192.168.1.1) was not authorized to send emails for vitalcarehealthsolutions.com.
- **DKIM (DomainKeys Identified Mail): Fail**, The email signature could not be verified, suggesting tampering or forgery.
- **DMARC (Domain-based Message Authentication, Reporting, and Conformance): Fail**, Due to the failure of both SPF and DKIM, the email failed DMARC validation. ●
Sending IP: 192.168.1.1 (Private) – Private IP addresses cannot send emails over the internet, confirming the message was spoofed.
- **Metadata Discrepancy:** The Message-ID <1234567890@vitalcarehealthsolutions.com> appeared synthetically generated rather than following standard mail server naming conventions.

Fig 2: Email headers revealing failed SPF, DKIM, and DMARC



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Fig 3: Headers identified in MXToolBox

2.3. URL and Link Analysis

The email contained a “Verify My Account Now” link that was linked to <http://malicious-link.com>, a domain that is not affiliated with VitalCare’s cloud infrastructure.

- **Static Link Analysis with VirusTotal:** Analysis of the embedded link using VirusTotal showed that **5 out of 95 security vendors** flagged the URL as malicious or associated with phishing activity, including ADMINUSLabs, alphaMountain.ai, CyRadar, Forcepoint ThreatSeeker, and Fortinet. Although the majority of vendors classified the link as clean, this does not guarantee safety, as newly created or obfuscated phishing domains can evade detection.

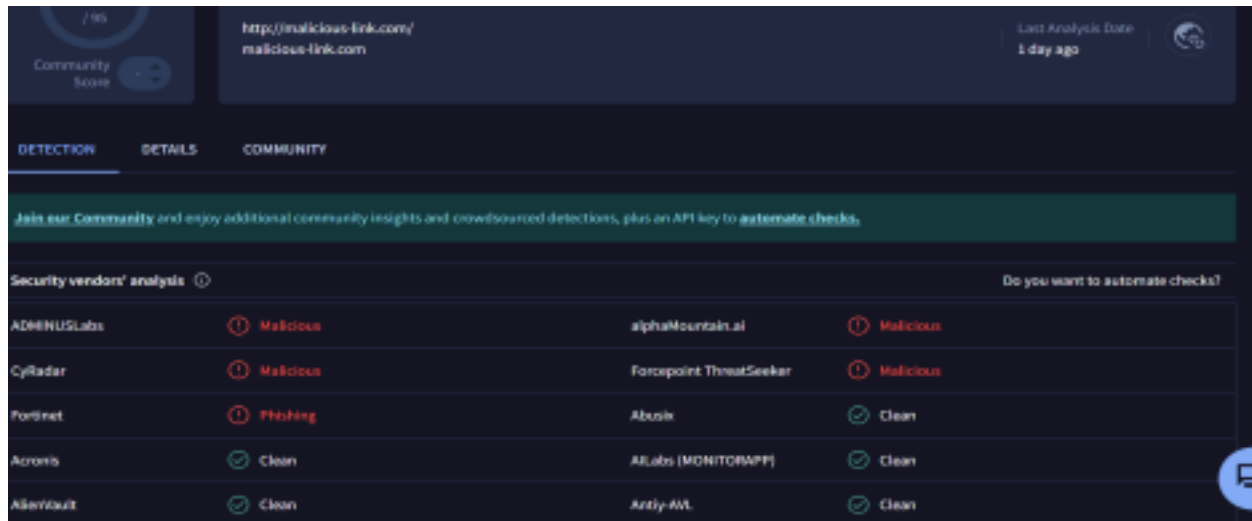


Fig 4: URL analysed in VirusTotal

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- **Dynamic link analysis with Browserling:** To further investigate the flagged link, the URL was safely opened within the **Browserling** sandbox environment. Upon accessing the link, the page redirected to a domain unrelated to VitalCare Health Solutions, confirming that the URL was crafted for phishing purposes rather than legitimate business activity.



Fig 5: Link analysis with Browserling

3. Tools Used

- VirusTotal: Scanned the embedded URL to check if it is flagged as malicious.
- MXToolbox Email Header Analyzer

- **Outlook:** Used to open the suspicious email and extract the raw header information to examine sender details, Message-ID, IP address, and authentication results (SPF, DKIM, DMARC).
- **Browserling:** Used to safely interact with the embedded URL in a sandboxed environment to observe how the link behaved (redirects, unrelated domain, phishing intent) without risking compromise to your system.

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4. Remediation and Eradication

Following the analysis, the following recommendations were made:

- **Multi-Channel Verification:** Implement a mandatory policy that all “Urgent” account actions must be verified via a secondary, independent communication channel (e.g., a verified internal chat or direct call) before clicking links.
- **Dynamic Analysis Integration:** Forward all suspicious attachments and URLs to a sandbox environment like Cuckoo Sandbox or FireEye for automated behavioral analysis.
- **Strict DMARC Policy:** Transition the DMARC record policies from NONE , which allows all emails including the ones that have failed authentications, to REJECT to ensure that unauthorized emails failing authentication are automatically dropped by receiving servers.
- The phishing URL (<http://malicious-link.com>) should be added to the organization's web filters and blocklists to prevent access. The sending IP (192.168.1.1) was a private address and does not represent a real internet-facing host, its presence instead confirms the email headers were spoofed, and should be flagged as an indicator of forged headers rather than an address to blocklist.
- **Report suspicious emails immediately:** All potentially harmful emails should be forwarded to the security team for analysis.
- **Maintain strong authentication:** Implement multi-factor authentication (MFA) for all critical accounts to ensure that even if credentials are compromised, attackers cannot gain

access.

- **Employee awareness and training:** Conduct regular cybersecurity awareness sessions focused on phishing recognition, safe email practices, and reporting procedures.

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5. Conclusion

In conclusion, the sophisticated nature of the phishing attempt targeting VitalCare Health Solutions explains the critical intersection of technology and human factors in the healthcare sector. While the organization's foundational expertise in providing secure, cloud-based EHR and telemedicine platforms is robust, the evolving landscape of phishing necessitates a need for dynamic, multi-layered defense.

The analysis of the email from *security@vitalcarehealthsolutions.com* confirms it was a phishing attempt. Header authentication failures (SPF, DKIM, DMARC), a private sending IP address, and a malicious link flagged by multiple security vendors all point to a deliberate attempt to steal user credentials. No compromise occurred thanks to timely identification and user caution. Ongoing monitoring, user awareness, and strong email security policies are essential to prevent similar incidents in the future.

This project underscores the necessity of a multi-layered defense—integrating persistent monitoring and threat intelligence with comprehensive user education. By aligning technical safeguards with proactive security protocols, organizations can effectively neutralize social engineering threats and ensure the integrity of their most sensitive data.

